修士論文

Microlocal Morse Theory and Truncated Microsupport
（超局所モース理論と打切り超局所台）

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1 Introduction

In this paper, We study microlocal Morse theory, which is presented, for instance, in [GKS12]. However, we replace the notion of microsupports with that of truncatd microsupprts, introduced by Kashiwara, Monteiro-Fernandes, and Schapira in [KFS03]. In [KFS03], they do not study the functorial property of truncated microsupport of direct images of non proper morphisms. We prove this functorial property under non proper direct images.

1.1 Organization

This paper is organized as follows. In Section **??**, we prove a truncated version of the noncharacteristic deformation lemma.

2 Notations and Reviews

We consider a commutative unital ring $\mathbf{k}$ of finite global dimension (e.g. $\mathbf{Z}$ or a field). Let X be a topological space. We denote by $\mathsf{D}^{\mathrm{b}}(\mathbf{k}_X)$ the bounded derived category of sheaves of $\mathbf{k}$-modules on X.

Let us denote Z a locally closed set of X and $i\colon Z \hookrightarrow X$ an inclusion. We use the following functors:

$$\mathrm{R}\Gamma_Z(\,\cdot\,) \coloneqq \mathrm{R}i_* i^!, \quad (\,\cdot\,)_Z \coloneqq \mathrm{R}i_! i^{-1}.$$

If there is no risk, we will write $\mathbf{k}_Z$ rather than $\mathbf{k}_{X,Z}$.

Geometrical Notations

Let M be a real manifold of class C^∞. We denote by $\pi\colon T^*M \to M$ the cotangent bundle of M. If $f\colon M \to N$ is a morphism of manifolds, then we have the natural diagram.

$$\begin{tikzcd}
TM \arrow[r, "f'"] \arrow[dr, "\tau_M"'] & M \underset{N}{\times} TN \arrow[r, "f_\tau"] \arrow[d] \arrow[dr, phantom, "\square"] & TN \arrow[d, "\tau_N"] \\
& M \arrow[r, "f"'] & N,
\end{tikzcd}
\qquad
\begin{tikzcd}
T^*M \arrow[dr, "\pi_M"'] & M \underset{N}{\times} T^*N \arrow[l, "f_d"'] \arrow[r, "f_\pi"] \arrow[d] \arrow[dr, phantom, "\square"] & T^*N \arrow[d, "\pi_N"] \\
& M \arrow[r, "f"'] & N.
\end{tikzcd}$$

In this diagram, f_τ and f_π denote the natural projections. f' is defined by $(x;v) \mapsto (x; df_x v)$, and f_d is induced by the transposition of f'.

3 Truncated Microsupport

Definition 3.1. Let $F \in \mathsf{D}^{\mathrm{b}}(\mathbf{k}_X)$. The closed conic subset $\mathrm{SS}_k(F)$ of T^*X is defined by: $p \notin \mathrm{SS}_k(F)$ if and only if there exists an open conic neighborhood U of p such that for any $x \in \pi(U)$ and for any real-valued C^1-function φ defined on a neighborhood of x such that $\varphi(x) = 0$, $d\varphi(x) \in U$, one has

(3.1) $$H^j_{\{\varphi \geqq 0\}}(F)_x = 0 \quad \text{for any } j \leqq k.$$

Functorial Properties

Proposition 3.2. Let $f\colon X \to Y$ be a morphism of real analytic manifolds and let $F \in \mathsf{D}^{\mathrm{b}}(\mathbf{k}_X)$ such that f is proper on $\mathrm{supp}(F)$. Then for any $k \in \mathbf{Z}$,

(3.2) $$\mathrm{SS}_k(\mathrm{R}f_*F) \subset f_\pi f_d^{-1}(\mathrm{SS}_k(F)).$$

The equality holds in case f is a closed embedding.

4 Lemmas from [KS90]

Consider a projective system of abelian groups $(M_n, \rho_{n,p})_n$ indexed by $\mathbf{N}$. One says that this system satisfies the Mittag-Leffler condition (ML for short) if for any $n \in \mathbf{N}$, the decreasing sequence $(\rho_{n,p}(M_p))_p$ of subgroups of M_n is stationary.

For a projective system of abelian groups $(M_n)_n$, we set

$$M_\infty \coloneqq \varprojlim_n M_n.$$

Consider a projective system of complexes

$$E_n^\bullet\colon \cdots \to M_n^{j-1} \to M_n^j \to M_n^{j+1} \to \cdots, \tag{4.1}$$

and projective limit

$$E_\infty^\bullet\colon \cdots \to M_\infty^{j-1} \to M_\infty^j \to M_\infty^{j+1} \to \cdots. \tag{4.2}$$

Denote by

$$\Phi_k\colon H^k(E_\infty^\bullet) \to \varprojlim_n H^k(E_n^\bullet) \tag{4.3}$$

the natural morphism.

Proposition 4.1 ([KS90, Proposition 1.12.4]). Assume that for each $j \in \mathbf{Z}$, the system $\big(M_n^{j-1}\big)_n$ satisfies the ML condition. Then

(a) for each $k \in \mathbf{Z}$, the map Φ_k is surjective,
(b) if moreover, for a given i the system $\big(H^{i-1}(E_n^\bullet)\big)_n$ satisfies the ML condition, then Φ_i is bijective.

The following is called Kashiwara's lemma.

Proposition 4.2 ([KS90, Proposition 1.12.6]). Let $(X_s, \rho_{s,t}\colon X_t \to X_s)_{(s,(s\leqq t))}$ be a projective system of sets indexed by $\mathbf{R}$. Assume that for each $s \in \mathbf{R}$, the canonical maps

$$\lambda_s\colon X_s \to \varprojlim_{r<s} X_r$$

and

$$\mu_s\colon \varinjlim_{t>s} X_t \to X_s$$

are both injective (resp. surjective). Then all the maps ρ_{s_0,s_1} $(s_0 \leqq s_1)$ are injective (resp. surjective).

ML procedure for sheaves

Proposition 4.3 ([KS90, Proposition 2.7.1]). Let X be a topological space and let $F \in \mathsf{D}^+(\mathbf{Z}_X)$. Let $(U_n)_{n\in\mathbf{N}}$ be an increasing sequence of open subsets of X, $(Z_n)_{n\in\mathbf{N}}$ a decreasing sequence of closed subsets of X. Set $U := \bigcup_n U_n$, $Z := \bigcap_n Z_n$.

(i) For any j, the natural map $\varphi_j\colon H_Z^j(U;F) \to \varprojlim_n H_{Z_n}^j(U_n;F)$ is surjective.
(ii) Assume that for a given j, the projective system $(H_{Z_n}^{j-1}(U_n;F))_n$ satisfies the ML condition. Then φ_j is bijective.
(iii) Let now $(X_n)_{n\in\mathbf{N}}$ be an increasing family of subsets of X satisfying $X = \bigcup_n X_n$ and $X_n \subset \mathrm{Int}(X_{n+1})$ for all n. Assume that for a given j, the projective system $\big(H^{j-1}(X_n;F)\big)_n$ satisfies the ML condition. Then the natural map $H^j(X;F) \to \varprojlim_n H^j(X_n;F)$ is bijective.

5 Truncated Noncharacteristic Deformation Lemma

Proposition 5.1. Let X be a Hausdorff space. Let $F \in \mathsf{D}^+(\mathbf{k}_X)$, and $(U_t)_{t\in\mathbf{R}}$ be a family of open subsets of X. Let $k \in \mathbf{Z}$. If s is a real number, set $Z_s := \bigcap_{t>s} \overline{U_t - U_s}$. We assume the following conditions.

(i) $U_t = \bigcup_{s<t} U_s$, for all $t \in \mathbf{R}$.

(ii) For all pairs (s,t) with $s \leqq t$, the set $\overline{U_t - U_s} \cap \operatorname{supp} F$ is compact.

(iii) For all $s \in \mathbf{R}$ and all $x \in Z_s - U_s$, we have

$$(H^j_{X-U_s}(F))_x = 0 \quad \text{for all } j \leqq k.$$

(iv) For all pairs (s,t) with $s < t$, $Z_s \subset U_t$.

Then, for all $t \in \mathbf{R}$, we have the isomorphisms

$$H^j\left(\bigcup_{s\in\mathbf{R}} U_s; F\right) \xrightarrow{\sim} H^j\left(U_t; F\right) \quad \text{for all } j < k,$$

and the injections

$$H^k\left(\bigcup_{s\in\mathbf{R}} U_s; F\right) \hookrightarrow H^k\left(U_t; F\right).$$

Example 5.2. Let us give examples of $(U_t)_t$ satisfying the condition iv, which is not contained in the original version of the proposition ([KS90, Proposition 2.7.2]). In all the examples below, X is $\mathbf{R}^2$.

(i) For a real number t, define the open set U_t by

$$U_t := \begin{cases} \{(x,y) \in X;\ \|(x,y)\| < t\} & (t > 0), \\ \varnothing & (t \leqq 0). \end{cases}$$

(ii) For a real number t, define the function $\varphi_t\colon \mathbf{R} \to \mathbf{R}$ as follows. If $t \leqq 0$, set $\varphi_t(x) \equiv 0$. If $t > 0$, set

$$\varphi_t(x) := \begin{cases} \sqrt{t^2 - x^2} & (|x| < s), \\ 0 & (|x| \geqq s). \end{cases}$$

Then define the open set U_t by

$$U_t := \{(x,y) \in X;\ y < \varphi_t(x)\}.$$

(iii) For a real number t, define the open set U_t as follows. If $t \leqq 0$, set

$$U_t := \{(x,y) \in X;\ y < 0\}.$$

If $t > 0$, set

$$U_t := \left\{ (x,y) \in X;\ \begin{array}{ll} \text{If } |x| < \frac{1}{t}, & y < t. \\ \text{Otherwise,} & y < \frac{1}{|x|}. \end{array} \right\}.$$

These families are visualized as Figure **??**.

Proof. Consider the following conditions for $j \in \mathbf{Z}$ and $s \in \mathbf{R}$:

$(a)^j_s$ $$\mu^j_s\colon \varinjlim_{t>s} H^j(U_t;F) \xrightarrow{\sim} H^j(U_s;F),$$

$(a')^k_s$ $$\mu^k_s\colon \varinjlim_{t>s} H^k(U_t;F) \hookrightarrow H^k(U_s;F),$$

$(b)^j_s$ $$\lambda^j_s\colon H^j(U_s;F) \xrightarrow{\sim} \varprojlim_{r<s} H^j(U_r;F).$$

Let us prove, for all $s \in \mathbf{R}$, $(a)^j_s$ for $j < k$ and $(a')^k_s$. We may assume that $\overline{U_t - U_s}$ is compact for any $s \leqq t$ by replacing X by $\operatorname{supp} F$. Consider the distinguished triangle

$$\mathrm{R}\Gamma_{X-U_s}(F) \to F \to \mathrm{R}\Gamma_{U_s}(F) \xrightarrow{+1}.$$

Let us endow with $Z_s \cup U_s$ the induced topology, and denote by $i_s\colon Z_s \cup U_s \hookrightarrow X$ the natural continuous map. Applying the functor

$$(\ \cdot\ )|_{Z_s\cup U_s} = i_s^{-1}$$

to the d.t. above, we have

$$\mathrm{R}\Gamma_{X-U_s}(F)|_{Z_s\cup U_s} \to F|_{Z_s\cup U_s} \to \mathrm{R}\Gamma_{U_s}(F)|_{Z_s\cup U_s} \xrightarrow{+1}.$$

Again applying to this triangle the functor $\mathrm{R}\Gamma(Z_s \cup U_s;\ \cdot\ )$, we have

(5.1)
$$\begin{tikzcd}
 & \mathrm{R}\Gamma(Z_s \cup U_s; \mathrm{R}\Gamma_{X-U_s}(F)|_{Z_s\cup U_s}) \arrow[dr] & \\
\mathrm{R}\Gamma(U_s;F) \arrow[ur, "+1"] & & \mathrm{R}\Gamma(Z_s \cup U_s; F|_{Z_s\cup U_s}). \arrow[ll]
\end{tikzcd}$$

By hypotheses (iii) and the definition of local cohomology, it follows that

$$H^j_{X-U_s}(F)\Big|_{Z_s\cup U_s} = 0 \quad \text{for any } j \leqq k,$$

and then

$$H^j\left(Z_s \cup U_s; \mathrm{R}\Gamma_{X-U_s}(F)|_{Z_s\cup U_s}\right) = 0 \quad \text{for any } j \leqq k.$$

Thus we have

$$H^j(Z_s \cup U_s; F|_{Z_s\cup U_s}) \xrightarrow{\sim} H^j(U_s;F) \quad \text{for any } j < k, \tag{5.2}$$
$$H^k(Z_s \cup U_s; F|_{Z_s\cup U_s}) \hookrightarrow H^k(U_s;F). \tag{5.3}$$

Therefore, if we show Lemma 5.3 below, we have

$$\begin{aligned}\varinjlim_{t>s} H^j(U_t;F) &\cong H^j(Z_s \cup U_s; F|_{Z_s\cup U_s}) \\ &\xrightarrow{\sim} H^j(U_s;F)\end{aligned}$$

for any $j < k$, and $(a)^j_s$ follows. Similarly, we have

$$\begin{aligned}\varinjlim_{t>s} H^k(U_t;F) &\cong H^k(Z_s \cup U_s; F|_{Z_s\cup U_s}) \\ &\hookrightarrow H^k(U_s;F)\end{aligned}$$

and $(a')^k_s$ follows.

Lemma 5.3. For any $j \in \mathbf{Z}$ and $s \in \mathbf{R}$, there exists a natural isomorphism

$$\varinjlim_{t>s} H^j(U_t; F) \cong H^j(Z_s \cup U_s; F|_{Z_s \cup U_s}).$$

Proof of Lemma 5.3 First we prove that for $F \in \mathrm{Mod}(\mathbf{k}_X)$, the natural morphism

$$\varinjlim_{t>s} \Gamma(U_t; F) \xrightarrow{\sim} \Gamma(Z_s \cup U_s; F|_{Z_s \cup U_s}). \tag{5.4}$$

is isomorphic. Consider the following diagram.

$$\begin{tikzcd}
\varinjlim_{t>s} \Gamma(U_t; F) \arrow[d] \arrow[dr] & \\
\varinjlim_{U \supset Z_s} \Gamma(U \cup U_s; F) \arrow[r] \arrow[d] & \Gamma(Z_s \cup U_s; F|_{Z_s \cup U_s}) \arrow[d] \\
\varinjlim_{U \supset Z_s} \Gamma(U; F) \arrow[r] & \Gamma(Z_s; F|_{Z_s})
\end{tikzcd}$$

Note that $(U_t)_{t>s}$ is a cofinal family of $(U \cup U_s)_{U \supset Z_s}$. Indeed, かく.

Therefore, we have

$$\varinjlim_{t>s} \Gamma(U_t; F) \xrightarrow{\sim} \varinjlim_{U \supset Z_s} \Gamma(U \cup U_s; F).$$

By this isomorphism, it is enough to show that

$$\phi\colon \varinjlim_{U \supset Z_s} \Gamma(U \cup U_s; F) \to \Gamma(Z_s \cup U_s; F)$$

is isomorphic. Since X is Hausdorff and Z_s is compact, the natural morhism

$$\psi\colon \varinjlim_{U \supset Z_s} \Gamma(U; F) \to \Gamma(Z_s; F)$$

is isomorphic (see [KS90, Proposition 2.5.1]). Since U_s is open, we have another isomorphism

$$\psi'\colon \varinjlim_{V \supset U_s} \Gamma(V; F) \to \Gamma(U_s; F).$$

The injectivity of ϕ follows from the definition of a sheaf and the injectivities of ψ and ψ'.

Let us prove that ϕ is surjective. Let $\sigma \in \Gamma(Z_s; F)$ be a section of $F|_{Z_s \cup U_s}$. By the isomorphism ψ above, we can extend $\sigma|_{Z_s}$ to an open neighborhood U of Z_s in X. More precisely, there exists a section τ on U such that $\tau|_{Z_s} = \sigma|_{Z_s}$. Now we see that $\tau|_{Z_s} = (\tau|_{(Z_s \cup U_s) \cap U})\big|_{Z_s}$, and by the definition of a section on the closed subset Z_s of $(Z_s \cup U_s) \cap U$, we can find an open subset W of $(Z_s \cup U_s) \cap U$ such that

$$\tau|_W = \sigma|_W. \tag{5.5}$$

By the definition of the induced topology of $(Z_s \cup U_s) \cap U$, there exists an open set $\widetilde{W}$ of X such that

$$W = ((Z_s \cup U_s) \cap U) \cap \widetilde{W} \quad \text{and} \quad \widetilde{W} \subset U.$$

Thus we have $W = ((Z_s \cup U_s)) \cap \widetilde{W}$. For this $\widetilde{W}$, the pair $\left(Z_s \cup U_s, \widetilde{W}\right)$ is an open covering of $\widetilde{W} \cup U_s$. Thus by (5.5), we can glue σ and $\tau|_{\widetilde{W}}$ together to get a section $\widetilde{\sigma}$ such that

$$\widetilde{\sigma}|_{Z_s \cup U_s} = \sigma, \quad \widetilde{\sigma}|_{\widetilde{W}} = \tau|_{\widetilde{W}}.$$

This proves the surjectivity of ϕ.

Next Let $F \in \mathsf{D}^+(\mathbf{k}_X)$. Let $I^\bullet$ be a complex of flabby sheaves quasi-isomorphic to F. Then it follows from (5.4) that

$$\begin{aligned}
\varinjlim_{t>s} H^j(U_t; F) &\cong \varinjlim_{t>s} H^j\Gamma(U_t; I^\bullet) \\
&\cong H^j\left(\varinjlim_{t>s} \Gamma(U_t; I^\bullet)\right) \\
&\cong H^j\Gamma(Z_s \cup U_s;\, I^\bullet|_{Z_s \cup U_s}) \\
&\cong H^j(Z_s \cup U_s;\, F|_{Z_s \cup U_s}).
\end{aligned}$$

This completes the proof. □

Let us come back to the proof of Proposition 5.1. We prove $(b)^j_s$ for all $j \in \mathbf{Z}$ and $s \in \mathbf{R}$ by induction on j. Since $F \in \mathsf{D}^+(\mathbf{Z}_X)$, we have $(b)^j_s$ for $j \ll 0$ and $s \in \mathbf{R}$. Assume $(b)^j_s$ is proved for all $j < j_0 (< k)$ and $s \in \mathbf{R}$. Since we have $(a)^j_s$ and $(b)^j_s$ for each $j < j_0$ and each $s \in \mathbf{R}$, we can apply Kashiwara's lemma (Proposition 4.2) and have

$$H^j(U_t; F) \xrightarrow{\sim} H^j(U_s; F) \quad \text{for } j < j_0 \text{ and } s \leqq t.$$

Fix $s \in \mathbf{R}$ and consider the projective system $\left(H^{j_0 - 1}(U_{s-1/n})\right)_{n \in \mathbf{N}}$. This system satisfies ML condition, so, by applying Proposition 4.3 (iii), we have

$$H^j(U_s; F) \xrightarrow{\sim} \varprojlim_n H^j\left(U_{s-\frac{1}{n}}; F\right).$$

Since $(s - 1/n)_n$ is a cofinal family of $(t)_{s>t}$, we have $(b)^{j_0}_s$. Then by induction on j, we have $(b)^j_s$ for $j < k$ and $s \in \mathbf{R}$. Again by applying Proposition 4.3 to the projective system $\left(H^j(U_n; F)\right)_n$, we obtain

$$\begin{aligned}
H^j\left(\bigcup_{s \in \mathbf{R}} U_s; F\right) &\cong H^j\left(\bigcup_{n \in \mathbf{N}} U_n; F\right) \\
&\xrightarrow{\sim} \varprojlim_n H^j\left(U_n; F\right)
\end{aligned}$$

and for each n, we have by Kashiwara's lemma that

$$H^j\left(U_n; F\right) \cong H^j(U_s; F)$$

for $n \geqq s$.

Finally we replace $(a)^j_s$ by $(a')^k_s$ for $j = k$, and obtain

$$H^k\left(\bigcup_{s \in \mathbf{R}} U_s; F\right) \hookrightarrow H^j\left(U_s; F\right).$$

This completes the proof. □

6 Relative Microlocal Morse Theory via Truncated Microsupport

Let $f\colon Y \to X$ be a morphism of manifolds, and φ a real valued function of class C^1. For a real number $t \in \mathbf{R}$, we set

$$\begin{aligned} Y_t &\coloneqq \{y \in Y \,;\, \varphi(y) < t\} = \varphi^{-1}((-\infty, t)), \\ \widetilde{Y}_t &\coloneqq \{y \in Y \,;\, \varphi(y) \leqq t\} = \varphi^{-1}((-\infty, t]). \end{aligned}$$

Let us define the following morphisms:

$$\begin{tikzcd} Y_t \arrow[r, "j_t"] \arrow[dr, "f_t"'] & Y \arrow[d, "f"] \\ & X, \end{tikzcd} \qquad \begin{tikzcd} \widetilde{Y}_t \arrow[r, "\tilde{j}_t"] \arrow[dr, "f_t"'] & Y \arrow[d, "f"] \\ & X. \end{tikzcd}$$

Proposition 6.1. Let $G \in \mathsf{D}^{\mathrm{b}}(Y)$, and assume $f_t\colon \operatorname{supp}(G) \cap \overline{Y_t} \to X$ is proper for all $t \in \mathbf{R}$. Let k be an integer and $t_\circ$ a real number.

(i) Assume that, for any $y \in Y - \widetilde{Y}_{t_\circ}$, we have

$$d\varphi(y) \notin \left(\mathrm{SS}_k(G) + f_d\left(Y \underset{X}{\times} T^*X\right)\right).$$

Then

(a)	$\mathrm{R}^j f_* G \cong \mathrm{R}^j f_{t*} j_t^{-1} G \quad$ for $j < k$ and $t > t_\circ$,
(a_k)	$\mathrm{R}^k f_* G \hookrightarrow \mathrm{R}^k f_{t*} j_t^{-1} G \quad$ for $t > t_\circ$,
(a′)	$\mathrm{R}^j \tilde{f}_* G \cong \mathrm{R}^j \tilde{f}_{t*} \tilde{j}_t^{-1} G \quad$ for $j < k$ and $t \geqq t_\circ$,
(a'_k)	$\mathrm{R}^k \tilde{f}_* G \hookrightarrow \mathrm{R}^k \tilde{f}_{t*} \tilde{j}_t^{-1} G \quad$ for $t \geqq t_\circ$,
(b)	$\mathrm{SS}_k(\mathrm{R}f_* G) \subset f_\pi\left(f_d^{-1}\left(\mathrm{SS}_k(G) \cap \pi^{-1}(\widetilde{Y}_{t_\circ})\right)\right) = f_\pi f_d^{-1}\left(\mathrm{SS}_k(G)\right).$

(ii) Assume that, for any $y \in Y - \widetilde{Y}_{t_\circ}$, we have

$$-d\varphi(y) \notin \left(\mathrm{SS}_k(G) + f_d\left(Y \underset{X}{\times} T^*X\right)\right).$$

Then

(c)	$\mathrm{R}^j f_! G \cong \mathrm{R}^j f_{t!} j_t^{-1} G \quad$ for $j < k$ and $t > t_\circ$,
(c_k)	$\mathrm{R}^k f_! G \hookrightarrow \mathrm{R}^k f_{t!} j_t^{-1} G \quad$ for $t > t_\circ$,
(c′)	$\mathrm{R}^j \tilde{f}_! G \cong \mathrm{R}^j \tilde{f}_{t*} \tilde{j}_t^{!} G \quad$ for $j < k$ and $t \geqq t_\circ$,
(c'_k)	$\mathrm{R}^k \tilde{f}_! G \hookrightarrow \mathrm{R}^k \tilde{f}_{t*} \tilde{j}_t^{!} G \quad$ for $t \geqq t_\circ$,
(d)	$\mathrm{SS}_k(\mathrm{R}f_! G) \subset f_\pi\left(f_d^{-1}\left(\mathrm{SS}_k(G) \cap \pi^{-1}(\widetilde{Y}_{t_\circ})\right)\right) \subset f_\pi f_d^{-1}\left(\mathrm{SS}_k(G)\right).$

Proof. Let $\tilde{f}$ be the morphism $(f,\varphi)\colon Y \to X \times \mathbf{R}$, and let q and $\tilde{\varphi}$ be the first and second projections defined on $X \times \mathbf{R}$ respectively. Then $\tilde{f}$ is proper on $\mathrm{supp}(G)$ and $f = q \circ \tilde{f}$. This can be proved as follows. If K is a compact set of $X \times \mathbf{R}$, $\tilde{f}^{-1}(K)$ is compact.

Let i_t be the open embedding $X \times (-\infty, t) \hookrightarrow X \times \mathbf{R}$ and $\tilde{i}_t$ the closed embedding $X \times (-\infty, t] \hookrightarrow X \times \mathbf{R}$. Set $q_t = q \circ i_t$ and $\tilde{q}_t = q \circ \tilde{i}_t$. These morphisms can be visualized by the following diagrams.

$$\begin{tikzcd}
Y \arrow[r, "\tilde{f}"] & X \times \mathbf{R} \arrow[r, "q"] & X, \\
Y_t \arrow[u, hook, "j_t"] \arrow[r, "\hat{f}_t"] & X \times (-\infty, t) \arrow[u, hook, "i_t"'] \arrow[ur, "q_t"'] &
\end{tikzcd}
\qquad
\begin{tikzcd}
Y \arrow[r, "\tilde{f}"] & X \times \mathbf{R} \arrow[r, "q"] & X. \\
\widetilde{Y}_t \arrow[u, hook, "\tilde{j}_t"] \arrow[r, "\check{f}_t"] & X \times (-\infty, t] \arrow[u, hook, "\tilde{i}_t"'] \arrow[ur, "\tilde{q}_t"'] &
\end{tikzcd}$$

Then, setting $\widetilde{G} := \mathrm{R}f_* G$, we have

$$\begin{aligned}
\mathrm{R}f_*G &\cong \mathrm{R}q_*\widetilde{G}, & \mathrm{R}\tilde{f}_*G &\cong \mathrm{R}\tilde{q}_*\widetilde{G},\\
\mathrm{R}f_{t*}j_t^{-1}G &\cong \mathrm{R}q_{t*}i_t^{-1}\widetilde{G}, & \mathrm{R}\tilde{f}_{t*}\tilde{j}_t^{-1}G &\cong \mathrm{R}\tilde{q}_{t*}\tilde{i}_t^{-1}\widetilde{G},\\
\mathrm{R}f_{t!}j_t^{-1}G &\cong \mathrm{R}q_{t!}i_t^{-1}\widetilde{G}, & \mathrm{R}\tilde{f}_{t*}\tilde{j}_t^{!}G &\cong \mathrm{R}\tilde{q}_{t*}\tilde{i}_t^{!}\widetilde{G}.
\end{aligned}$$

Thus taking j-th cohomology, we have

$$\begin{aligned}
\mathrm{R}^jf_*G &\cong \mathrm{R}^jq_*\widetilde{G}, & \mathrm{R}^j\tilde{f}_*G &\cong \mathrm{R}^j\tilde{q}_*\widetilde{G},\\
\mathrm{R}^jf_{t*}j_t^{-1}G &\cong \mathrm{R}^jq_{t*}i_t^{-1}\widetilde{G}, & \mathrm{R}^j\tilde{f}_{t*}\tilde{j}_t^{-1}G &\cong \mathrm{R}^j\tilde{q}_{t*}\tilde{i}_t^{-1}\widetilde{G},\\
\mathrm{R}^jf_{t!}j_t^{-1}G &\cong \mathrm{R}^jq_{t!}i_t^{-1}\widetilde{G}, & \mathrm{R}^j\tilde{f}_{t*}\tilde{j}_t^{!}G &\cong \mathrm{R}^j\tilde{q}_{t*}\tilde{i}_t^{!}\widetilde{G}.
\end{aligned}$$

Therefore, it is enough to prove the result for $\widetilde{G}$, q, i_t, $\tilde{\varphi}$ instead of G, f, j_t, φ.

(i) It follows from the hypothesis that

$$\mathrm{SS}_k(\widetilde{G}) \cap (T^*X \times \{(t;\tau); t > t_\circ\}) \subset T^*X \times \{(t;\tau); \tau \leqq 0\}\,. \tag{6.1}$$

Indeed, by hypothesis, for any $(y;\eta) \in \mathrm{SS}_k(G)$ and $(f(x);\xi) \in Y \times_X T^*X$, we have

$$d\varphi(y) \neq \eta + f^*\xi. \tag{6.2}$$

If $(x,t;\xi,\tau)$ is a point of $\mathrm{SS}_k(\widetilde{G}) \cap (T^*X \times \{(t;\tau); t > t_\circ\})$, then, by the formula (3.2), there exists $(y;\eta) \in \mathrm{SS}_k(G)$ such that

$$\tilde{f}(y) = (x,t) \text{ and } \eta = \tilde{f}^*(\xi,\tau).$$

For this η, we have

$$\eta = \tilde{f}^*(\xi,\tau) = f^*\xi + \varphi^*\tau = f^*\xi + \tau d\varphi(y).$$

If $\tau > 0$, we get

$$d\varphi(y) = \frac{1}{\tau}\eta - \frac{1}{\tau}f^*\xi$$

and this contradicts (6.2) because $\mathrm{SS}_k(G)$ is conic.

By (6.1), Let U be an open subset of X contained in a local chart, and convex in such a chart. By Proposition **??**, we have isomorphisms for all $t > t_\circ$ and $j < k$

$$H^j(U \times \mathbf{R}; \widetilde{G}) \xrightarrow{\sim} H^j(U \times (-\infty, t)\,; \widetilde{G}) \tag{6.3}$$

and have an injection for all $t > t_\circ$

$$H^k(U \times \mathbf{R}; \widetilde{G}) \hookrightarrow H^k(U \times (-\infty, t)\,; \widetilde{G}). \tag{6.4}$$

This can be showed as follows.

This proves the result. □

Remark 6.2. $(U \times I_t)$ is not a gentle family.

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